

APS360 PROJECT PROPOSAL: GENERATING SONG LYRICS WITH ARTIFICIAL INTELLIGENCE

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1 INTRODUCTION

Traditional songwriting is a complex artistic process that requires the writer to not only create lyrics but also take into account melodies and musical structure in order to create a song that can express emotions, tell stories, or convey messages. For many, this process is also halted by “writer’s block,” in which the writer feels stuck and uninspired for an undetermined amount of time.

In an effort to shorten these periods of writers block and alleviate the pressure on writers, the proposed project aims to develop a deep learning model capable of generating song lyrics. By training a model on a large array of existing music, the outcome aims to be a co-creative partner to songwriters across the industry.

This project is interesting as it aims to combine the logic of deep learning with the unstructured creativity of art. Whilst tasks like computation, coding, and classification are rigid with deterministic outcomes, human art lacks any rhyme or reason, unable to be limited by programmable rules. This project goes against the nature of AI, but should it succeed, it will provide a way for a computer program to mimic the fluidity of human creativity and serve as a starting point for songwriters facing writers block.

Deep learning is the most reasonable approach for this task as lyric generation falls under the umbrella of natural language processing (NLP), a field saturated with deep learning techniques [1]. Deep learning relies on complex and abstract patterns, such as songwriting’s use of metaphor and rhyme, which are difficult to define with a basic set of rules. The model will need to learn through the use of a vast set of existing song lyrics in order to internalize the nuances of language and musical style.

2 ILLUSTRATION OF OVERALL MODEL

The illustration of the RNN/transformer model for lyrics generation.

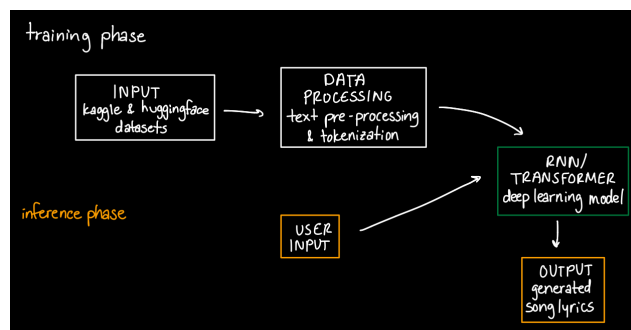


Figure 1: Illustration of lyrics generation model

3 BACKGROUND

Automatic song lyric generation is a rapidly growing field at the corner of music information retrieval and NLP, focused on creating coherent, stylistically sound lyrics based on prompts, music genres, or melodies [2][3]. This task is challenging because models must handle rhyme schemes, rhythm and syllable counts, genre-specific vocabulary, and musical phrase structure while still maintaining semantic coherence [4].

Earlier systems relied on RNNs trained as language models on artist or genre specific lyrical datasets [2]. LSTM-based models have been able to reproduce genre specific stylistic properties such as line length, word diversity, and characteristic phrasing for rap and pop lyrics, demonstrating that a sequence models can capture nontrivial stylistic behavior in lyrics [2]. Similar LSTM-RNN lyric generators implemented in open-source projects train character or word level models on scraped lyrics to imitate the style of particular artists or genres [5].

More recent work integrates musical context and more advanced architectures. Melody-conditioned lyric generators use LSTM networks that take both previous words and melodic features such as pitch, beat position, and velocity as input, enabling the model to align generated lyrics with musical phrasing and rhythm [6]. Hierarchical and transformer-based approaches, such as unsupervised melody-to-lyric generation, introduce higher-level song structure and explicit syllable and rhythm constraints, yielding lyrics that are more singable and coherent than simpler RNN baselines. End-to-end conditional rap lyric generators further refine transformer-style models on rap datasets to better capture rhyme density and stylistic coherence in specific subgenres like rap.

4 DATA PROCESSING

This project will use two lyrical datasets from HuggingFace [7][8] and one dataset from Kaggle [9]. The HuggingFace collection contains roughly 180k MIDI files with aligned UTF-8 text lyric files, including a deduplicated subset of about 47k unique song pairs that will be used to limit repetition and noise. The Kaggle dataset provides a large set of Genius lyrics with automatically inferred language labels, which will be filtered for English lyrics and used as additional text-only training data.

For preprocessing, first the songs will be indexed by matching MIDI and lyric filenames, then examples with missing, empty, or very short lyric files will be discarded. Lyric text will be lowercased and stripped of markup such as section headers or chord tags such that all datasets share the same formatting. Finally, songs will be split into non-overlapping training, validation, and test sets at the song level to prevent different versions of the same song from appearing across splits.

5 ARCHITECTURE

The project will utilize an RNN with Long Short-Term Memory (LSTM) cells due to their effectiveness in modelling sequential text data. LSTMs are specifically chosen to allow the model to "selectively retain or forget information over long periods," which is essential for maintaining thematic consistency across song lyrics [2]. Furthermore, the project may explore a Transformer architecture, which uses self-attention mechanisms to process text in parallel. This enables the system to capture intricate stylistic patterns and long-range dependencies more efficiently than standard sequential models, providing a sophisticated framework for creative text generation.

6 BASELINE MODEL

N-gram language models have been widely used as simple baselines for lyric generation in prior work, including standalone lyrics generators [10]. In this project, the baseline will be an n-gram language model that estimates the probability of each next word from the frequencies of short word sequences in the training sets and then samples words sequentially to form lines. An n-gram count table will be constructed from the lyrics, smoothing will be applied, and at generation time, the model will repeatedly choose the next word given the previous one or two words until a line-length limit is reached. This non-neural, melody-agnostic model is simple to implement and reproduce,

and it provides a clear reference point for assessing whether the transformer-based model improves coherence, diversity, and stylistic consistency.

7 ETHICAL CONSIDERATIONS

Some of the considerations surrounding this project include copyright integrity and the potential for bias within the algorithm. Since the model is trained on existing lyrical datasets, it may generate output that mimics copyrighted material, raising concerns regarding plagiarism and the intellectual property of original artists. Furthermore, the training data also presents a limitation in cultural diversity; since the selected datasets are skewed toward Western popular music, the model may struggle to represent or accurately reproduce the structural nuances of non-Western songwriting traditions and diverse linguistic styles.

8 LINK TO GITHUB

<https://github.com/jasjaapdua/aps360-project>

9 CITATIONS, FIGURES, TABLES, REFERENCES

- [1] “Deep Learning for Song Lyrics Generation,” arxiv.org. [Online]. Available: <https://arxiv.org/pdf/1807.10854>
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- [10] “Lyrics Generation using N-Gram Technology,” ijert.org. [Online]. Available: <https://www.ijert.org/research/lyrics-generation-using-n-gram-technology-IJERTV2IS80652.pdf>